

DOE SBIR/STTR FY26 PHASE I • GENESIS MISSION

Topic 1, Scaling the Biotechnology Revolution

Copy each answer into the matching open-text field in AMP. Word counts are shown so you can check them against the Pitch Development Guide before pasting. Images and video are not enabled, so all four answers are plain text.

TOTAL WORDS	QUESTION 1	TOPIC
690 across four answers	95 held under 100	Locked after the pitch is submitted

PROJECT TITLE TO ENTER IN AMP

Physics-Informed Machine Learning for Design of Continuous Immobilized-Enzyme Reactors
Converting Post-Consumer PET to Fuel-Grade Ethanol

01

Summary, Topic, and Mission Alignment

95 WORDS

PlastiBioFuel runs two loops: a structured wet lab converting post-consumer PET into fuel-grade ethanol, and a fast simulation loop that decides what the lab runs next. The chemistry is a PET hydrolase immobilized in a custom covalent organic framework inside a continuous spiral-flow reactor. Scale-up is limited by design search, not enzyme performance. Phase I hardens the physics-informed machine learning model we already run against our own continuous-operation data and closes the loop: the model proposes conditions, the reactor tests them, results retrain the model. That is AI for bioreactor design and biomanufacturing, Topic 1.

02

Technical Promise

199 WORDS

Significance. Enzymatic PET recycling is limited by reactor engineering, not enzyme discovery. Conditions are still chosen one factor at a time, days per run, so almost none of the design space gets searched.

Innovation. FAST-PETase was reported by the Alper group at UT Austin in 2022, itself produced by machine learning. PlastiBioFuel did not invent the enzyme. Our contributions, covered by USPTO provisional 63/848,456, are the framework immobilization system, the spiral-flow geometry, and the integrated

depolymerization-to-ethanol process. Phase I moves machine learning up one level, from residue selection to reactor operation: the surrogate predicts conversion and turnover frequency from temperature, residence time, crystallinity, and loading, and an active-learning loop selects the next run.

Feasibility. Both halves already exist. On the bench we have depolymerized high-crystallinity post-consumer bottle PET to terephthalic acid, MHET, and ethylene glycol, confirmed by HPLC, loaded the framework to near-complete occupancy with activity retained, held product formation linear through 24 hours, and shown turnover-frequency decline tracking substrate depletion, not enzyme deactivation. In parallel we have run a coupled multi-scale reactor model across more than a million simulations and checked its predictions against that measured output. Phase I hardens it into an active-learning loop and tests it on held-out runs.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 34 WORDS

Phase I will produce a model whose predictions are tested against held-out reactor runs, so the go or no-go evidence at the end of Phase I is a measured prediction error, not a claim.

03 Commercialization Potential

199 WORDS

Value proposition. The product is fuel-grade ethanol made from a waste feedstock. Post-consumer PET is abundant, domestically sourced, priced well below agricultural feedstock, and it does not compete with food production. Low carbon fuel standard credit value applies to the finished fuel and is the firmer part of the value case. Federal renewable identification number eligibility remains subject to final renewable-fuel pathway qualification, and we do not assume it.

Competitive advantage. Pyrolysis carries high energy cost per gallon; mechanical recycling degrades polymer on every cycle. Enzymatic recyclers that return monomer to virgin-equivalent resin sell into the resin market, so our offtake channel does not overlap theirs. AI-for-bioprocess software vendors have to contract for the asset we own: a working reactor generating the training data. Wet-lab groups search one condition at a time; we search at software speed. Owning both loops is what shortens the bench-to-plant step, where enzymatic routes usually stall.

Go to market. We are in offtake discussions with ARG Petro, a regional fuel distributor, and have their interest. Texas puts PET waste aggregation, refining infrastructure, and fuel blending demand in one region. The Phase I output is a sizing and operating envelope for the first modular unit, so the model feeds a capital decision.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 23 WORDS

A first modular unit sited near a Texas PET aggregation point is the specific commercial target that Phase I is meant to de-risk.

04 Team Qualifications

197 WORDS

Company. Michael David Ward, Founder and CEO, is the principal investigator and is primarily employed by PlastiBioFuel. He built the project's simulation and modeling capability and runs it himself, so the computational work Phase I extends stays inside the company rather than moving to a subcontractor. PlastiBioFuel is a service-disabled veteran-owned small business verified through SBA VetCert and holds USPTO provisional 63/848,456. Its laboratory program produced the preliminary data cited here. All work will be performed in the United States.

Partnerships. The University of North Texas is our research partner under executed Sponsored Research Agreement SRA1016, with a \$250,000 extension under negotiation. Dr. Shenggqian Ma, Department of Chemistry, leads covalent organic framework and immobilization chemistry; his group has published on enzyme immobilization in covalent organic frameworks. Dr. Joshua Phipps, postdoctoral researcher, runs day-to-day wet lab work. The partnership gives Phase I reactor access, HPLC analytics, and materials characterization with no capital spending.

Follow-on funding. PlastiBioFuel has a separate NSF SBIR Phase I proposal, Research.gov 328095, submitting against the November 4, 2026 window. That proposal covers wet chemistry and reactor hardware and does not overlap the computational scope proposed here. We disclose it so DOE can confirm there is no duplicate funding.

OPTIONAL ADD-ON, ONLY IF THE GUIDE ALLOWS MORE WORDS · 19 WORDS

Marshall Nadel is a seed investor in PlastiBioFuel, and Patrick Tarlton of the Texas Concrete Association supports business development.

WORD COUNT SUMMARY

As written · With add-on

01. Summary, Topic, and Mission Alignment	95 words as written · 95 with add-on
02. Technical Promise	199 words as written · 233 with add-on
03. Commercialization Potential	199 words as written · 222 with add-on
04. Team Qualifications	197 words as written · 216 with add-on
Total	690 words as written · 766 with add-on

Question 1 is held under 100 words. If the Pitch Development Guide sets higher limits, append the add-on sentences. If it sets lower limits, the answers cut cleanly at paragraph boundaries. Drop the Feasibility paragraph from Question 2 last, since it carries the preliminary data that makes the pitch credible.

OPTIONAL UPLOAD: BIBLIOGRAPHY AND REFERENCES CITED

Prepared and included in this package. Worth uploading: it substantiates the statement that PlastiBioFuel did not invent the enzyme, and it shows the UNT partner's published record in enzyme immobilization in covalent organic frameworks.

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